DSP in VLSI

Homework 3

Interpolator

Student ID

M11407439

Name

MING-HONG, LIN

Development Environment

HDL	Verilog HDL
Simulation Tools	Synopsys VCS and Verdi
Synthesis Tool	Synopsys DC Compiler
Process	TSMC 16nm(ADFP)
Verification	Python

Due Date: 2026/04/15

Contents

1 Step 1 — Interpolator Function Verification **2**
1.1 Result 1 - Linear vs Second-Order Polynomial Interpolator 2

2 Step 2 — Interpolator Function Verification **4**
2.1 Result 2 - Linear vs Piecewise Parabolic Interpolator 4

3 Step 3 — Bit-True Model (BF16) **5**
3.1 Result 3 - BF16 Operation Results 5

4 Step 4 — Performance Degradation by Finite Precision **6**
4.1 Result 4 - Quantization Error Analysis 6

5 Step 5 — Hardware Reduction **7**
5.1 Result 5 - Original Block Diagram 7
5.2 Result 5 - Hardware Sharing Approaches 8

6 Step 6 — Implementation **9**
6.1 Result 6 - Behavior Simulation & Error Analysis 9
6.2 Result 7 - Timing Report and Constraint 10
6.3 Result 8 - Post-Synthesis Simulation 12
6.4 Result 8 - BF16 Bit-True Model vs. Verilog Output 13

Step 1 — Interpolator Function Verification

1.1 Result 1 - Linear vs Second-Order Polynomial Interpolator

Assume that

$$x_1[m] = \cos\left(2\pi\left(\frac{mT_s}{10T_s} + \frac{\phi}{2}\right)\right) + j\sin\left(2\pi\left(\frac{mT_s}{10T_s} + \frac{\phi}{2}\right)\right),$$

where $\phi = \mathrm{mod}(d, 2)$ and d is the last digit of your student ID. The sampling frequency is 10 times the frequency of the sinusoidal wave. Please use **linear interpolator** and **second-order polynomial interpolator** to interpolate the sampled waveform in the region of $10 \le m \le 20$ with $\mu = 0, \frac{1}{8}, \frac{2}{8}, \ldots \frac{7}{8}$. Note that you have $x_1[9]$ when you want to interpolate $x_1[10.125]$. Show the **real-part** and **imaginary-part** output waveform $\hat{x}_1[m+\mu]$ after interpolation for both methods. In addition, draw the absolute error $|x_1[m+\mu] - \hat{x}_1[m+\mu]|$ in the region of $10 \le m \le 20$ with $\mu = 0, \frac{1}{8}, \frac{2}{8}, \ldots \frac{7}{8}$. (30%)

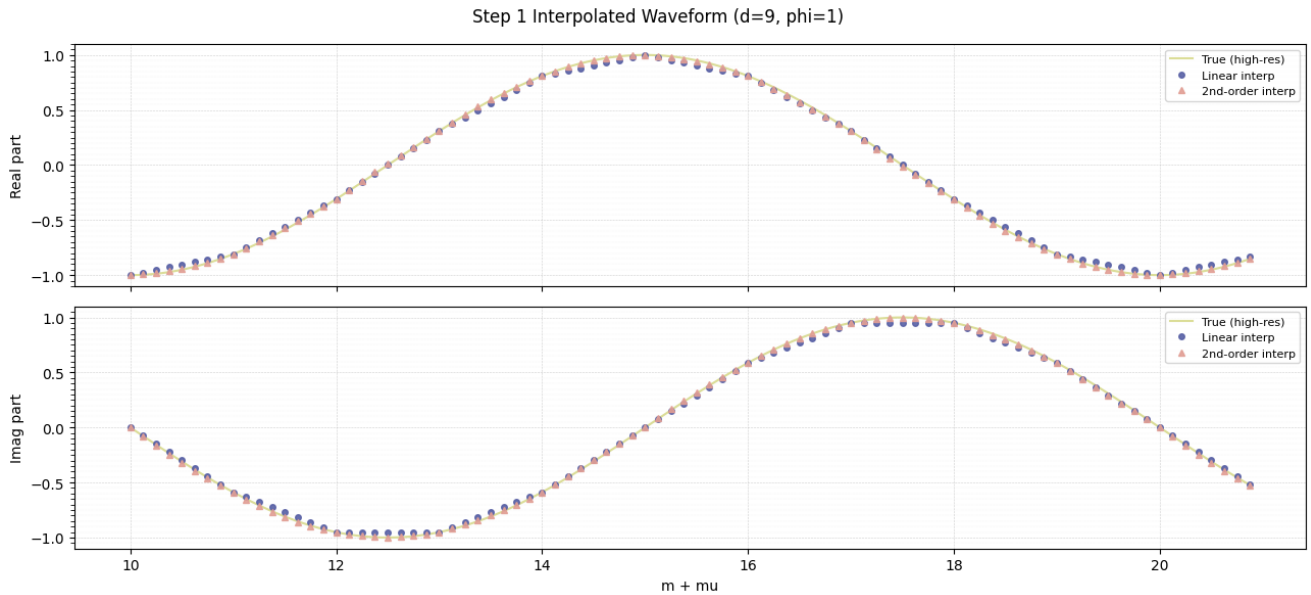

Figure 1.1: Waveform of Linear vs Second-Order Polynomial Interpolator

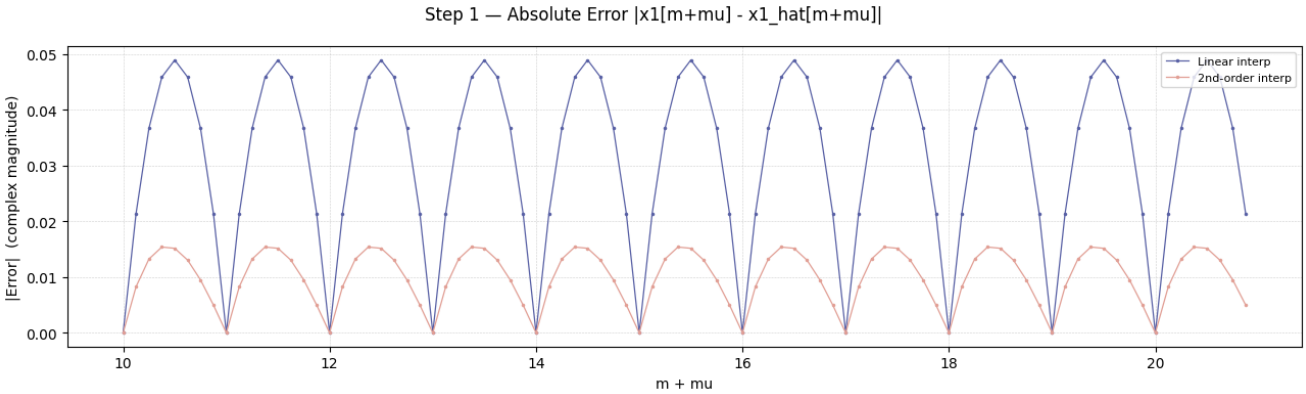

Figure 1.2: Absolute Error Comparison of Linear vs Second-Order Polynomial Interpolator

Step 2 — Interpolator Function Verification

2.1 Result 2 - Linear vs Piecewise Parabolic Interpolator

Assume that

$$x_2[m] = 2^{-20}\left[\cos\left(2\pi\left(\frac{mT_s}{4T_s} + \frac{\phi}{3}\right)\right) + j\sin\left(2\pi\left(\frac{mT_s}{4T_s} + \frac{\phi}{3}\right)\right)\right],$$

where ϕ is the same parameter according to your student ID. The sampling frequency is 4 times the frequency of the sinusoidal wave. Please use **linear interpolator** and **piecewise parabolic interpolator** to interpolate the sampled waveform in the region of $4 \le m \le 8$ with $\mu = 0, \frac{1}{9}, \frac{2}{9}, \ldots \frac{8}{9}$. Show the **real-part** and **imaginary-part** output waveform $\hat{x}_2[m+\mu]$ after interpolation. In addition, draw the absolute error $|x_2[m+\mu] - \hat{x}_2[m+\mu]|$ in the region of $4 \le m \le 8$ with$\mu = 0, \frac{1}{8}, \frac{2}{8}, \ldots \frac{7}{8}$. (30%)

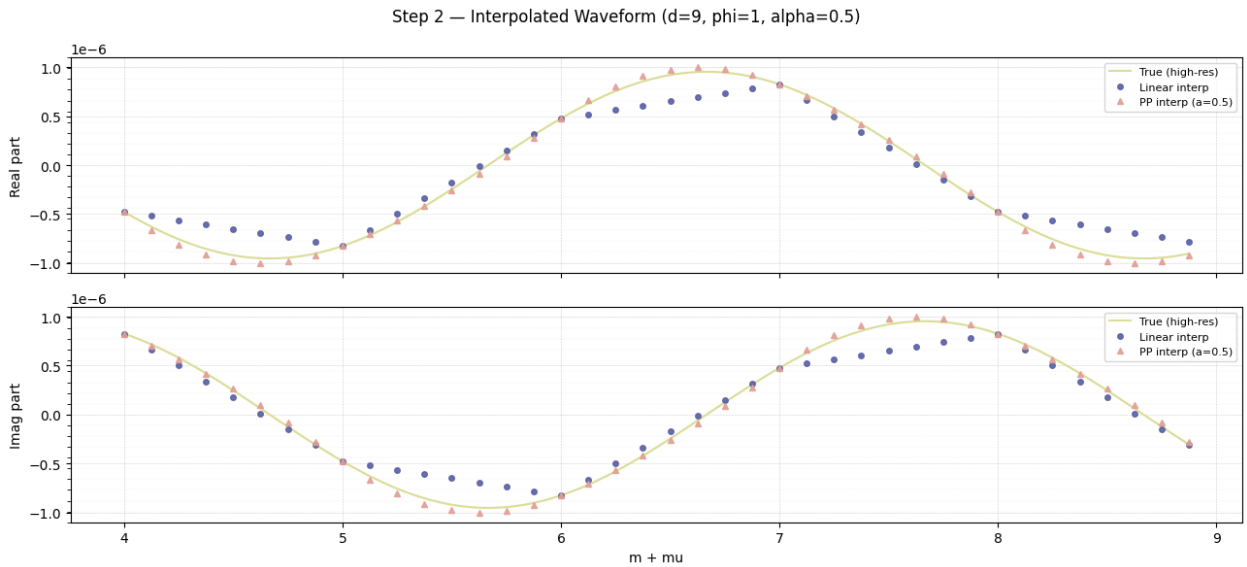

Figure 2.1: Waveform of Linear vs Piecewise Parabolic Interpolator

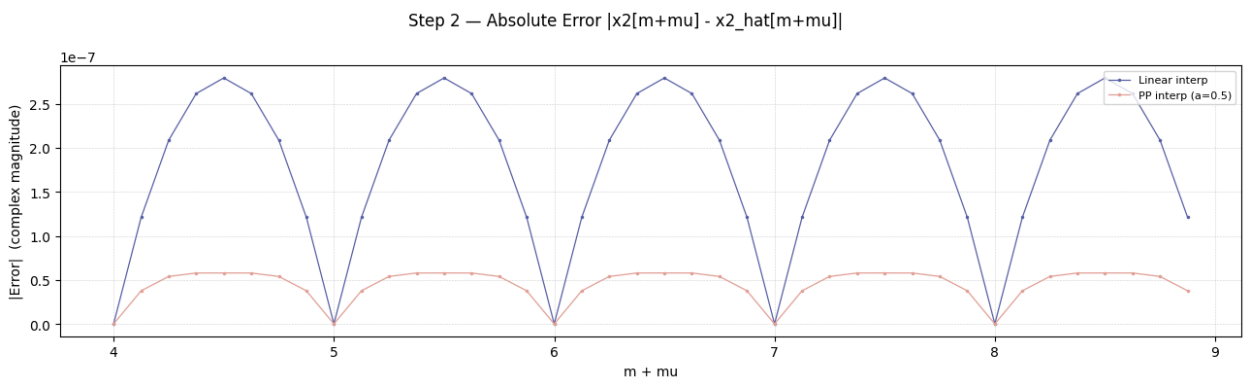

Figure 2.2: Absolute Error Comparison of Linear vs Piecewise Parabolic Interpolator

Step 3 — Bit-True Model (BF16)

3.1 Result 3 - BF16 Operation Results

Please use **BF16 format** to realize addition and multiplication. Write the bit-true model (simulated program in Python) to realize BF16 additions and multiplication. For simplicity, NaN, infinity, and subnormal numbers are not handled — if the magnitude is smaller than the smallest normal number, use **truncation** to make it become 0. Truncation is also used to avoid renormalization. Show the results calculated by your BF16 bit-true model for the following operands. Express results both in **decimal** and **binary** representation (S = sign bit, E = exponent field, F = fraction field, all in binary). **(20%)**

	Operand 1	**Operand 2**	**Operation**
(a)	S: 1 E: `1000_0011` F: `0000_000`	S: 0 E: `1000_1011` F: `0000_011`	Addition
(b)	S: 1 E: `0000_0001` F: `0000_011`	S: 0 E: `0000_0001` F: `1111_010`	Addition
(c)	S: 1 E: `0000_0010` F: `1100_000`	S: 0 E: `0111_1100` F: `0000_110`	Multiplication
(d)	S: 0 E: `0110_0011` F: `1011_110`	S: 0 E: `1001_0011` F: `0101_000`	Multiplication

Table 3.1: BF16 operands and operations.

The following are the four option results obtained from a BF16 Bit-True Model implemented in Python.

	Operation	**Decimal**	**S**	**E & F (binary)**
(a)	Addition	4192	0	E: `1000_1011`, F: `0000_011`
(b)	Addition	0	0	E: `0000_0000`, F: `0000_000`
(c)	Multiplication	0	0	E: `0000_0000`, F: `0000_000`
(d)	Multiplication	0.0088501	0	E: `0111_1000`, F: `0010_001`

Table 3.2: BF16 operation results.

Step 4 — Performance Degradation by Finite Precision

4.1 Result 4 - Quantization Error Analysis

Please analyze the error of the **second-order interpolator** if its arithmetic operations are changed to BF16 by using the bit-true model. Draw the figures indicating the difference of interpolated results caused by data format for outputs $\hat{x}_1[m+\mu]$ in the region of $10 \le m \le 20$ with $\mu = 0, \frac{1}{8}, \frac{2}{8}, \ldots \frac{7}{8}$. **(10%)**

As shown in the figure below, switching the arithmetic from double precision to BF16 introduces a noticeable quantisation error due to the reduced mantissa width (7 bits) of the BF16 format. The peak absolute errors observed over the interpolation range $10 \le m \le 20$ are summarised below:

Component	**Max Absolute Error**
Real part	1.5959×10^{-2}
Imaginary part	1.6048×10^{-2}

Table 4.1: Maximum absolute error introduced by BF16 arithmetic in the second-order interpolator.

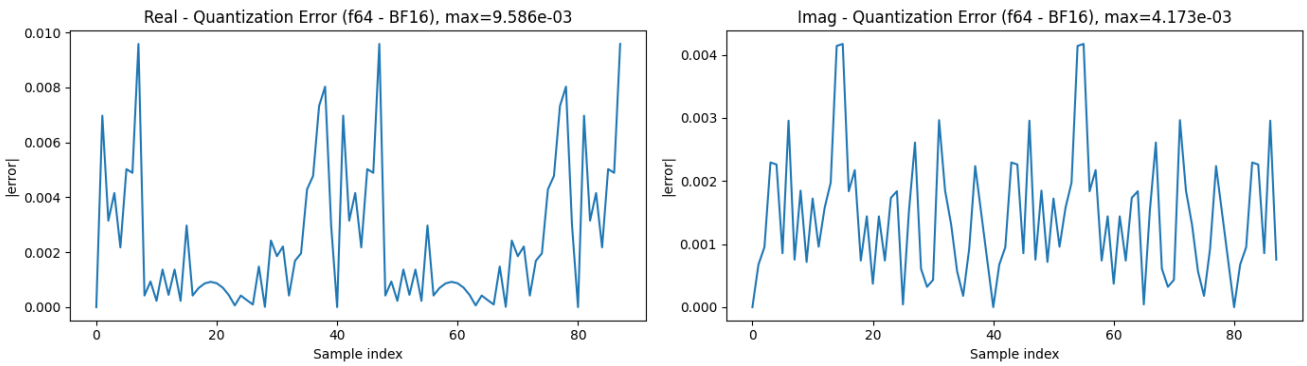

Figure 4.1: Interpolation Error Caused by BF16 Finite Precision (Second-Order Interpolator)

Step 5 — Hardware Reduction

5.1 Result 5 - Original Block Diagram

Draw the complete block diagram of your first version implementation according to Fig. 4. (5%)

The second-order polynomial interpolator in Farrow structure decomposes the interpolation into three fixed sub-filters v_0, v_1, and v_2, whose outputs are combined as

$$\hat{x}[m+\mu] = v_0 + \mu\, v_1 + \mu^2\, v_2.$$

The coefficients of each sub-filter are derived from three consecutive input samples $X[m]$, $X[m+1]$, and $X[m+2]$ as summarised in Table 5.1.

Sub-filter	$X[m]$	$X[m+1]$	$X[m+2]$
v_2	$+\frac{1}{2}$	-1	$+\frac{1}{2}$
v_1	$-\frac{3}{2}$	$+2$	$-\frac{1}{2}$
v_0	$+1$	0	0

Table 5.1: Farrow sub-filter coefficients for the second-order polynomial interpolator.

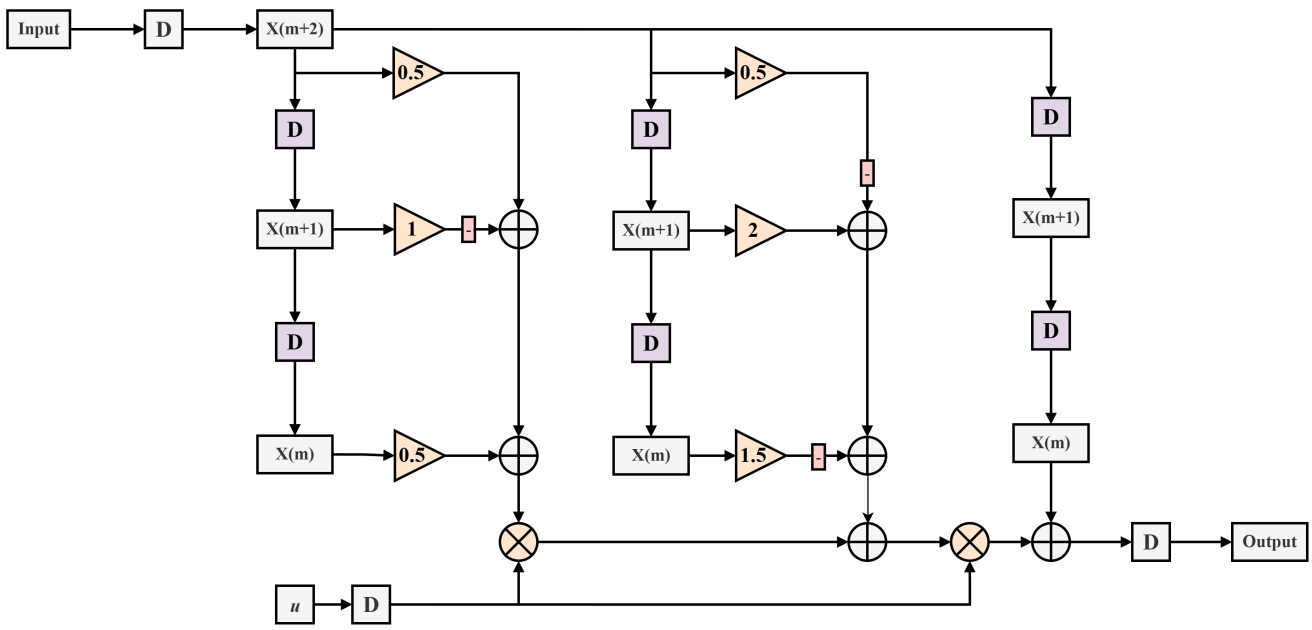

Figure 5.1: Original Block Diagram (Farrow Structure)

5.2 Result 5 - Hardware Sharing Approaches

Show your hardware sharing version (The precision of the datapath must be BF16 and can not be changed.) (10%) Draw the block diagram after your improvement. (5%) Indicate the critical path (5%)

Three hardware optimizations were applied to reduce resource usage and improve operating frequency.

1. **Shared delay-line (single DFF chain).** The original structure uses three independent delay lines, one per sub-filter branch. Since all three branches consume the same input samples $X[m]$, $X[m+1]$, and $X[m+2]$, a single shared shift-register chain suffices, eliminating two redundant sets of D flip-flops.

2. **Multiplier-free scaling.** The dominant coefficient in the design is 0.5. In BF16 format, multiplication by 0.5 reduces to decrementing the exponent field by one and no multiplier is needed. Accordingly, all $\times 0.5$ operations are replaced by a one-bit exponent adjustment. The coefficient $\times 2$ is likewise converted to an equivalent addition ($A + A$), and $\times 1.5$ is decomposed into $A + 0.5A$, so that the entire datapath requires no general-purpose multipliers.

3. **Two-stage pipeline insertion.** To increase the maximum operating frequency, two pipeline registers are inserted across the critical path, balancing the combinational logic depth between stages.

As we can see, the critical path is the purple path, which includes one multiplier and one adder.

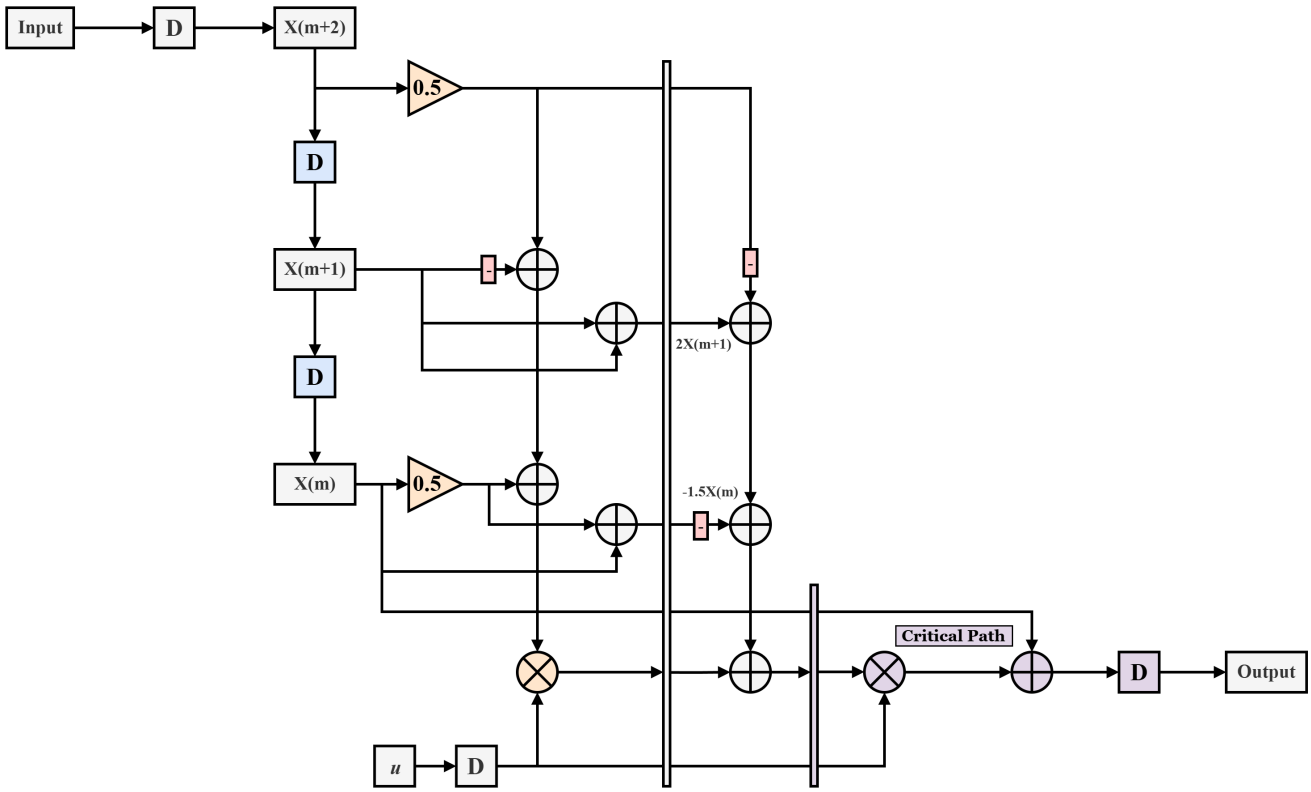

Figure 5.2: Block Diagram with Hardware Sharing Approaches

Step 6 — Implementation

6.1 Result 6 - Behavior Simulation & Error Analysis

Show the timing diagram of behavior simulation for the **first 8** and **last 8** ($\mu = 0, \frac{1}{8}, \ldots \frac{7}{8}$) interpolated outputs of $\hat{x}_1[m + \mu]$ in the region of $10 \leq m \leq 20$. Show the **real-part** and **imaginary-part** error of hardware implementation results compared to the double-precision results. **(30%)**

Due to the two-stage pipeline architecture and the delayed feedback loops (DFF) at the input and output stages, the design introduces a latency of 4 clock cycles between input and output. As shown in Figure 6.2, the output results are fully presented 4 clock cycles after the input signal Mu stops to enter the system.

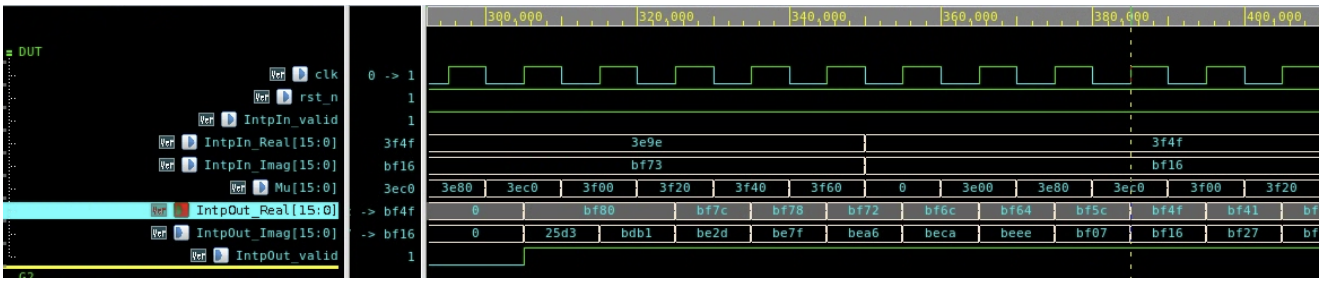

Figure 6.1: Timing diagram of behavior simulation for the first 8

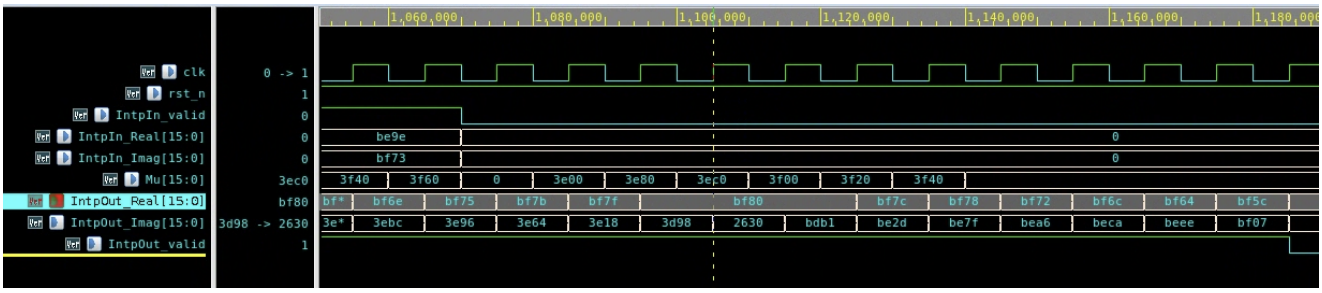

Figure 6.2: Timing diagram of behavior simulation for the last 8

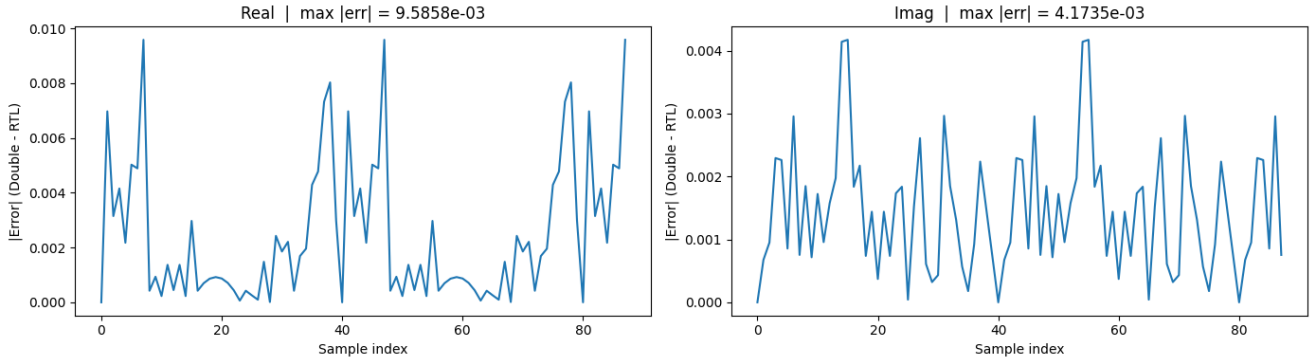

Figure 6.3: Error Analysis of hardware results compared to the double-precision results

6.2 Result 7 - Timing Report and Constraint

how your **timing report** to verify the critical path of the block diagram in Step 5. **(10%)** List your **timing constraint** for post-synthesis simulation. **(5%)**

Parameter	Value (ns)
Clock Period	1
Clock Uncertainty	0.1
Clock Latency	0.1
Input / Output Maximum Delay	0.5

Table 6.1: Timing constraint settings provided by TA

According to the Timing Report, the critical path starts from **uV2_V1_Temp** Register and ends at **IntpOut_Real** Register, which is consistent with the purple path depicted in Section 5.2.

```
Operating Conditions: ff0p88v125c   Library: N16ADFP_StdCellff0p88v125c_ccs
Wire Load Model Mode: segmented

  Startpoint: Real_Part/uV2_V1_TEMP_reg[7]
              (rising edge-triggered flip-flop clocked by clk)
  Endpoint: IntpOut_Real_reg[13]
            (rising edge-triggered flip-flop clocked by clk)
  Path Group: clk
  Path Type: max

  Des/Clust/Port     Wire Load Model       Library
  ------------------------------------------------
  Interpolator       ZeroWireload          N16ADFP_StdCellff0p88v125c_ccs
  Common_Logic_0     ZeroWireload          N16ADFP_StdCellff0p88v125c_ccs

  Point                                                    Incr       Path
  -----------------------------------------------------------------------
  clock clk (rise edge)                                    0.00       0.00
  clock network delay (ideal)                              0.50       0.50
  Real_Part/uV2_V1_TEMP_reg[7]/CP (DFCNQD1BWP20P90)        0.00       0.50 r
  Real_Part/uV2_V1_TEMP_reg[7]/Q (DFCNQD1BWP20P90)         0.04       0.54 f
  U2694/Z (OR2D1BWP20P90)                                  0.02       0.56 f
  U3168/CO (FA1D1BWP20P90)                                 0.03       0.58 f
  U3169/CO (FA1D1BWP20P90)                                 0.03       0.61 f
```

Figure 6.4: Report_Timing 1

As demonstrated in the report, the design achieves a maximum operating frequency of 1 GHz without any setup and hold time violations.

```
U915/ZN (NR2D1BWP16P90)                                  0.01       1.25 f
U1324/ZN (IND2D2BWP16P90)                                0.02       1.27 f
U3681/ZN (NR2D1BWP16P90)                                 0.02       1.29 r
U1315/ZN (MAOI222D1BWP16P90)                             0.02       1.31 f
U3693/ZN (AOI21D1BWP16P90)                               0.01       1.32 r
U3694/ZN (ND2D1BWP16P90)                                 0.01       1.33 f
U3702/ZN (NR2D1BWP16P90)                                 0.01       1.35 r
U1285/ZN (INVD1BWP16P90)                                 0.01       1.35 f
U3703/ZN (NR2D1BWP16P90)                                 0.01       1.36 r
U1252/ZN (ND2D1BWP16P90)                                 0.01       1.38 f
U3712/ZN (AOAI211D1BWP16P90)                             0.01       1.39 r
IntpOut_Real_reg[13]/D (DFCNQD1BWP16P90)                 0.00       1.39 r
data arrival time                                                   1.39

clock clk (rise edge)                                    1.00       1.00
clock network delay (ideal)                              0.50       1.50
clock uncertainty                                       -0.10       1.40
IntpOut_Real_reg[13]/CP (DFCNQD1BWP16P90)                0.00       1.40 r
library setup time                                      -0.01       1.39
data required time                                                  1.39
-------------------------------------------------------------------------
data required time                                                  1.39
data arrival time                                                  -1.39
-------------------------------------------------------------------------
slack (MET)                                                         0.00
```

Figure 6.5: Report_Timing 2

6.3 Result 8 - Post-Synthesis Simulation

Show the timing diagram of **post-synthesis simulation** for the first 8 and last 8 interpolated outputs of $\hat{x}_1[m+\mu]$ in the region of $10 \leq m \leq 20$. Show the **real-part** and **imaginary-part** error of post-synthesis hardware outputs compared to the double-precision results. **(20%)**

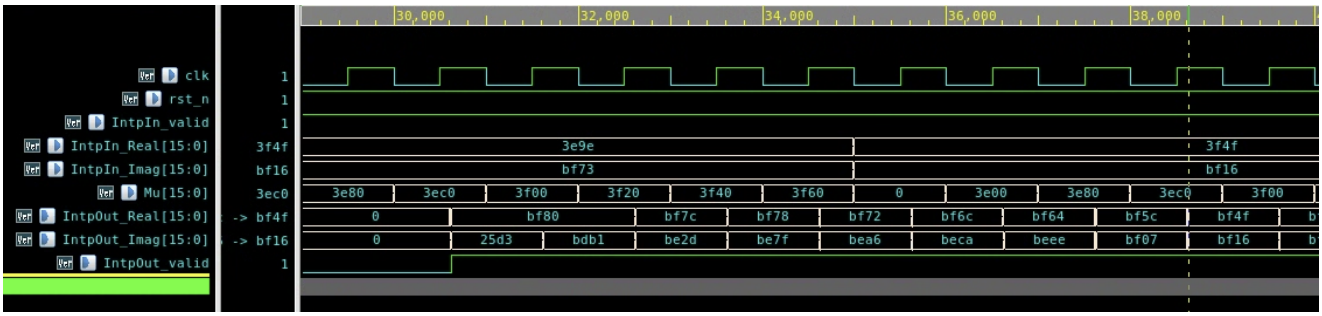

Figure 6.6: Timing diagram of post-synthesis simulation for the first 8

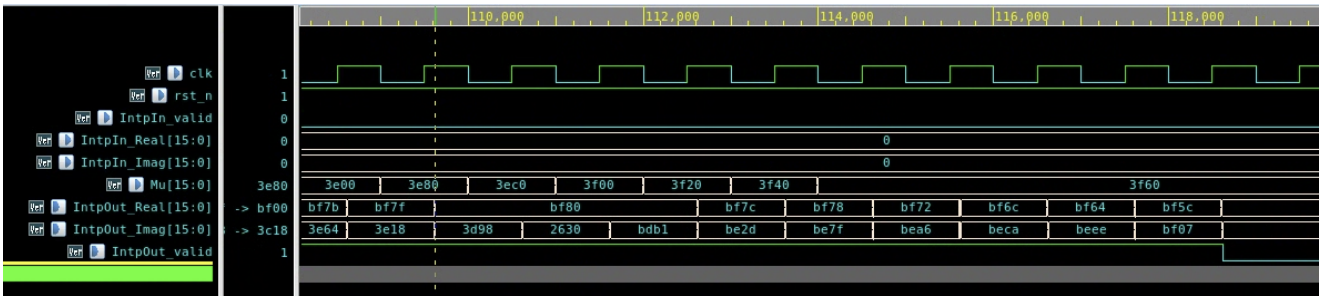

Figure 6.7: Timing diagram of post-synthesis simulation for the last 8

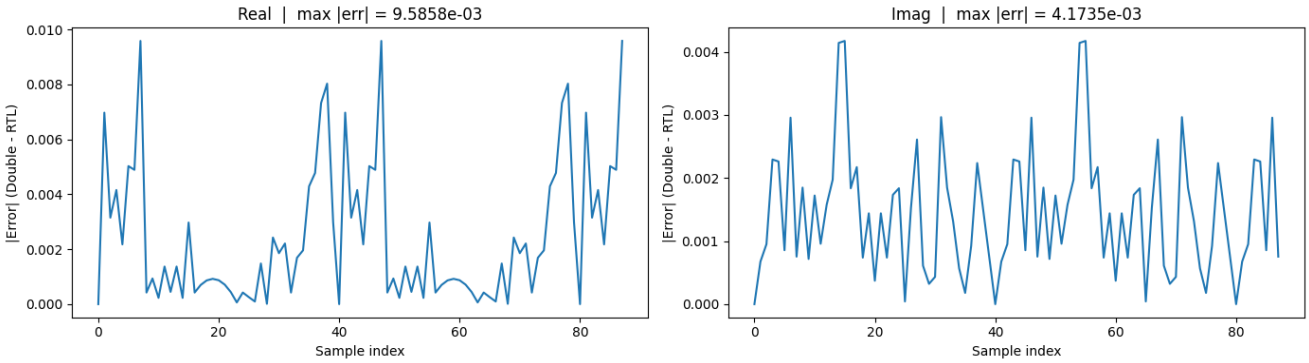

Figure 6.8: Error Analysis of hardware results compared to the double-precision results (Post-Synthesis)

6.4 Result 8 - BF16 Bit-True Model vs. Verilog Output

Construct the **BF16 bit-true model** following the architecture after hardware improvement. Depict the **real-part** and **imaginary-part** errors of interpolated outputs $\hat{x}_1[m+\mu]$ between the Verilog outputs and the bit-true model in the region of $10 \leq m \leq 20$ with $\mu = 0, \frac{1}{8}, \frac{2}{8}, \ldots \frac{7}{8}$. **(20%)**

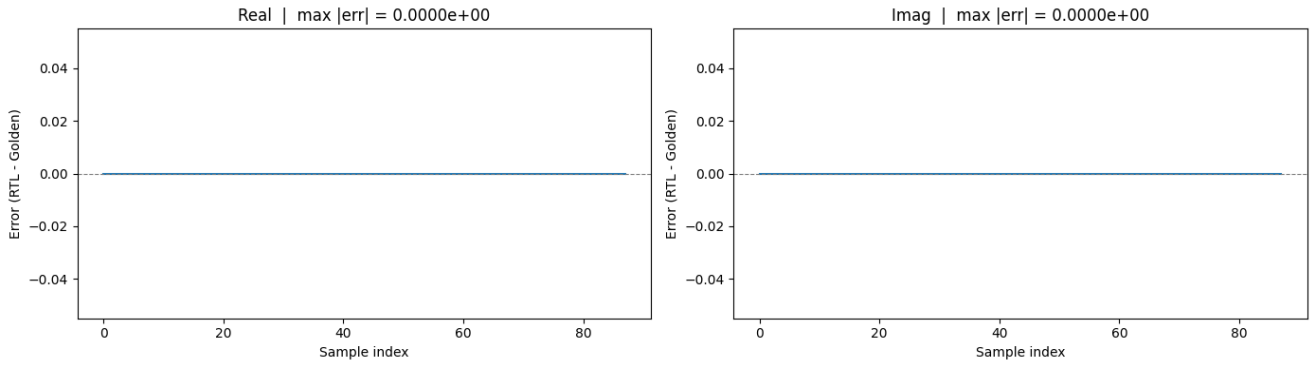

Figure 6.9: BF16 Bit-True Model vs. Verilog Output